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Specialization and interaction strength in a tropical plant–frugivore network differ among forest strata

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Abstract. The degree of interdependence and potential for shared coevolutionary history of frugivorous animals and fleshy-fruited plants are contentious topics. Recently, network analyses revealed that mutualistic relationships between fleshy-fruited plants and frugivores are mostly built upon generalized associations. However, little is known about the determinants of network structure, especially from tropical forests where plants' dependence on animal seed dispersal is particularly high. Here, we present an in-depth analysis of specialization and interaction strength in a plant–frugivore network from a Kenyan rain forest. We recorded fruit removal from 33 plant species in different forest strata (canopy, midstory, understory) and habitats (primary and secondary forest) with a standardized sampling design (3447 interactions in 924 observation hours). We classified the 88 frugivore species into guilds according to dietary specialization (14 obligate, 28 partial, 46 opportunistic frugivores) and forest dependence (50 forest species, 38 visitors). Overall, complementary specialization was similar to that in other plant–frugivore networks. However, the plant–frugivore interactions in the canopy stratum were less specialized than in the mid- and understory, whereas primary and secondary forest did not differ. Plant specialization on frugivores decreased with plant height, and obligate and partial frugivores were less specialized than opportunistic frugivores. The overall impact of a frugivore increased with the number of visits and the specialization on specific plants. Moreover, interaction strength of frugivores differed among forest strata. Obligate frugivores foraged in the canopy where fruit resources were abundant, whereas partial and opportunistic frugivores were more common on mid- and understory plants, respectively. We conclude that the vertical stratification of the frugivore community into obligate and opportunistic feeding guilds structures this plant–frugivore network. The canopy stratum comprises stronger links and generalized associations, whereas the lower strata are composed of weaker links and more specialized interactions. Our results suggest that seed-dispersal relationships of plants in lower forest strata are more prone to disruption than those of canopy trees.

Key words: Africa; birds; coevolution; diet specialization; frugivory; Kakamega Forest, Kenya; monkeys; mutualistic network; plant–animal interactions; rain forest; seed dispersal.

Introduction

Frugivory and animal seed dispersal are crucial processes in tropical rain forests (e.g., Cordeiro and Howe 2003, Sekercioglu 2006). In tropical forests, up to 90% of the plant species produce fleshy fruits that are eaten and dispersed by animals (Howe and Smallwood 1982, Jordano 2000). Populations of rain forest plants depend on seed dispersal because it provides gene flow among populations, circumvents distance- and density-dependent mortality of recruits, and enables the colonization of new sites (Nathan and Muller-Landau 2000). In turn, high proportions of animals, in particular birds, feed on fruits in the tropics (Kissling et al. 2009).

The degree to which plant–frugivore relationships are specialized, interdependent, and share coevolutionary history have been a matter of debate (Howe and Smallwood 1982, Levey and Benkman 1999, Loiselle et al. 2007). Although most plant–frugivore relationships are generalized, frugivores respond to plant traits such as fruit size and structure (Wheelwright 1985, Lomáscolo et al. 2008), and frugivore preferences shift in response to spatiotemporal variation in fruit abundance (Loiselle and Blake 1990, Carlo et al. 2003). Hence,

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diffuse nonequilibrium processes on the community rather than on species level are supposed to be the drivers of the evolution of plant–frugivore relationships (Herrera 1985, Wheelwright 1988).

In recent years, mutualistic relationships between plants and animals, such as plant–pollinator and plant–frugivore interactions, have been analyzed from a network perspective (Bascompte et al. 2003, Vázquez et al. 2009*a*) where the occurrence and frequency of interactions between plants and animal species are recorded on the community level. Most species in such networks have few and weak interactions, whereas a subset of core species exhibits many and strong interactions (Vázquez et al. 2009*a*). Moreover, interactions between mutualistic partners are typically asymmetric in their relative importance for the interacting species (Bascompte et al. 2003, Vázquez et al. 2007). These properties imply a nested structure of networks so that little-connected species interact with a subset of highly connected species (Bascompte et al. 2003), a configuration that is expected if commonly and rarely observed species interact randomly (Blüthgen et al. 2008). From these studies, we have gained insights into the structure of mutualistic networks, but we still know little about the determinants of network structure (Vázquez et al. 2009*b*).

The most important drivers of network structure are species abundance and resource availability, as well as complementarity of phenotypes, phenologies, spatial distributions, and phylogenetic relationships (Carnicer et al. 2009, Vázquez et al. 2009*a*, *b*). From an evolutionary perspective, trait complementarity and phylogenetic relationships set a blueprint for the realized interactions between pairs of species (species trait or forbidden link hypothesis; Thompson 2005, Stang et al. 2007) and determine the deviation from an expected probability distribution of interactions and thus the degree of complementary specialization within a network (Blüthgen et al. 2008, Blüthgen 2010). From an ecological perspective, species abundance and resource availability are important determinants of the number of interactions between two species and thus interaction strength (abundance or neutrality hypothesis; Jordano 1987, Vázquez et al. 2007). Studies that aim to identify determinants of specialization and interaction strength in mutualistic networks are still rare (but see Carnicer et al. 2009, Vázquez et al. 2009*b*), particularly from species-rich tropical systems. Moreover, very few studies have disentangled differential roles of functional guilds or higher taxonomic levels of plants and consumer species in the network (but see Olesen et al. 2007), although this is an important level of ecological organization, at least in plant–frugivore relationships (Herrera 1985).

Here, we present an in-depth analysis of complementary specialization and interaction strength in a plant–frugivore network from a Kenyan rain forest. We recorded fruit removal from plants in different forest strata (canopy, midstory, and understory) and habitats (primary and secondary forest), and classified frugivore visitors into guilds according to their niche breadth in terms of diet and habitat specialization. We expected that obligate frugivores feed on more fruit sources than opportunistic frugivores to meet their nutritional requirements (see Whelan et al. 1998), and that they forage mostly where fruit availability is high (Shanahan and Compton 2001, Saracco et al. 2004), such as in the forest canopy (Schaefer et al. 2002). Hence, from an evolutionary perspective, we hypothesized that obligate frugivores are less specialized in fruit choice than opportunistic frugivores (frugivory hypothesis), representing generalized plant–frugivore associations in the canopy (stratification hypothesis) and in habitats with plenty of fruits (habitat hypothesis). From an ecological perspective, we hypothesized that the impact of a frugivore species on plants increases with species abundance (abundance hypothesis) and specialization on specific plants (specialization hypothesis). Moreover, we tested whether interaction strength of different guilds of frugivores varied among strata and between habitats. We expected that the interaction strength of obligate frugivores is high in the forest canopy and that true forest species are more important for plants in primary than in secondary forest.

Methods

Study area

We conducted this study in a mid-altitudinal tropical rain forest in western Kenya (Kakamega Forest, 1500–1700 m a.s.l., 0°14′–0°21′ N, 34°47′–34°48′ E) that is considered to be the easternmost remnant of the Congo basin rain forests (Kokwaro 1988). The main forest of Kakamega Forest covers an area of 8245 ha and consists of patches of primary and secondary forest (Bleher et al. 2006). The northern part of Kakamega Forest has been under protection of the Kenya Wildlife Service since 1985 (Mitchell et al. 2009) and experiences rather low levels of human disturbance (Bleher et al. 2006). Mean annual precipitation is 1960 mm (Isecheno Forest Station 1982–2005), with a rainy season from April to November and a short dry season from December to March, when most plant species produce fruits.

Frugivore observations

From December 2007 to March 2008, we studied plant species ($n = 33$) with fleshy fruits in an area of ~250 ha in northern Kakamega Forest. The study area included the easily accessible parts of the Kakamega National Reserve in the Buyangu region comprising a mosaic of primary and secondary forest vegetation, as typical for Kakamega Forest (Farwig et al. 2008). Local field assistants daily surveyed the study area for plants bearing ripe fleshy fruits; surveys were conducted along the numerous transects and paths cutting through the study area. For each plant species recorded during the study period, representative and spatially separated

individuals bearing numerous ripe fruits were selected. According to availability of fruiting individuals, we selected four ($n = 27$ species), three ($n = 5$ species), or two individuals ($n = 1$ species) of each species.

For each individual, we estimated plant height with a clinometer (Suunto PM-5/360 PC, Vantaa, Finland) and the number of fruits on the observation day (on a logarithmic scale). For each species, 20 ripe fruits were collected from one or two individuals per species. Fruit samples were taken with telescopic scissors by local field assistants and, if necessary, trees were climbed to reach ripe fruits. Fruits were weighed on the day of collection. Mean crop mass per plant species was calculated as the product of mean fruit number and mean fruit mass.

To record the frugivore species feeding on each plant species, five local ornithologists equipped with binoculars (Nikon ACTION VII 10×50 CF, Tokyo, Japan) recorded frugivore visits to each plant individual from 07:00 to 14:00 hours. Each plant was observed once by a single observer camouflaged with green mesh; different individuals of the same species were observed by different observers. For the six plant species where fewer than four individuals were available, observations were repeated on the same individual on another day to standardize the sampling effort to 28 h per plant species. During observations, only fruit-eating individuals were recorded and counted; both mammals and birds were determined to species level according to Kingdon (1997) and Zimmermann et al. (1998), respectively. Observation time was sufficient to record a representative sample of the frugivore community for each plant species (Flörchinger et al. 2010).

We also recorded handling of fruits by frugivores and distinguished among swallowing, pecking, and dropping fruits. Predation of seeds was not observed. Fruit handling was observed in almost 90% of frugivore visits. Of these visits, swallowing of fruits or seeds was recorded in 66%, pecking in 52%, and dropping in only 12% of frugivore–fruit interactions. Percentages do not add up to 100% because a single visitor handled fruits in different ways so that some fruits were swallowed, while others were pecked or dropped.

Plant community

Of the 33 plant species, 19 grew in primary and 14 in secondary forest (species list in Appendix). Most plant species occurred only in a single forest type; those species found in both habitats ($n = 4$ species) were classified to belong to the forest type where they were more abundant. Primary forests harbor a tree diversity of ~40 species/ha and have a closed canopy at >30 m height (Farwig et al. 2008). In contrast, secondary forests grow on sites that have been logged in the 1980s, are less diverse (~20 species/ha), and are composed mostly of early-successional species that form an open canopy at heights of ~20 m (Farwig et al. 2008). Bird species composition has been shown to differ between primary and secondary forest, although both habitats occur in intermingled patches (Farwig et al. 2008, 2009). Moreover, study plants grow in different strata of the forest. We classified forest strata according to plant height and recorded frugivores on 10 species in the understory (<3 m), 8 in the midstory (3–15 m), and 15 in the canopy of the forest (>15 m). Our design was not two-factorial because fruiting trees were not equally distributed over strata in the two forest habitats, so we separately tested main effects of stratum and habitat.

Frugivore community

Overall, we observed 3447 interactions between 33 plant and 88 frugivore species (Appendix). Plant species that attracted the most frugivores were *Ficus* cf. *lutea* ($n = 630$ visits during 28 hours), *Ficus* cf. *thonningii* (498), and *Prunus africana* (398). The most frequent frugivores on fruiting plants were Common Bulbul (*Pycnonotus barbatus*, $n = 586$ observations), blue monkey (*Cercopithecus mitis*, 292), and the migratory Blackcap (*Sylvia atricapilla*, 287).

According to their diet, we classified the 88 frugivore species (i.e., 83 birds, four monkeys, and one squirrel) into obligate, partial, and opportunistic frugivores. Obligate frugivores depend on fruits as the only major food item, partial frugivores also use other major food items, and opportunistic frugivores only occasionally feed on fruits as supplemental food (for details see Kissling et al. 2007). We classified 14 species as obligate, 28 as partial, and 46 as opportunistic frugivores (for birds, see classification by Kissling et al. [2007]; for mammals, see food items given in Cords [1986], Kingdon [1997], Fashing [2001], Kunz and Linsenmaier [2007]; see also Appendix). Moreover, we classified frugivores according to their dependence on forest into 50 true forest species and 38 forest visitors (for birds, see classification by Bennun et al. [1996]; for mammals, see habitat preference given in Kingdon [1997]; see also Appendix). Forest species almost exclusively reproduce in forests, whereas visitors usually reproduce in non-forest habitats (see Bennun et al. [1996]; forest species include forest specialists and generalists).

Network analysis

We calculated interaction frequencies of each frugivore species with each plant species to construct a quantitative interaction matrix (Appendix; see the Interaction Web Database, *available online*).[7] We defined interaction frequency as the number of fruit-eating individuals on a plant species independent of fruit handling. The total frequency of a frugivore species was defined as the number of visits of a frugivore species on all plant species, whereas the total visitation frequency from the plant's perspective was given by the number of all frugivore visits on a particular plant species. Thus, we

[7] ⟨http://www.nceas.ucsb.edu/interactionweb⟩

used the marginal totals of the interaction matrix to estimate total frequencies (Blüthgen et al. 2007).

To assess the degree of complementary specialization (in the sense of niche partitioning; Blüthgen 2010), we compared the observed frequency distribution of interactions with an expected probability distribution that assumes that all species interact with their partners in proportion to their observed frequency totals (Blüthgen et al. 2006, 2007). At the network level, we calculated the deviation from the expected probability distribution as the standardized two-dimensional entropy H_2', and at the species level as the standardized Kullback-Leibler distance (d') for each plant species i and for each frugivore species j (Blüthgen et al. 2006, 2007). To compare complementary specialization among networks in different forest strata and habitats, we classified plant species into their respective stratum (canopy, midstory, or understory) and into primary or secondary forest, whereas frugivore species could occur in all or in a subset of strata and habitats. We calculated the specialization index (H_2') for each stratum and habitat. To obtain estimates of sampling error of H_2' for each stratum and habitat, we randomly sampled interactions from the matrix of observed interactions with replacement, generated an interaction matrix with the same overall frequency of interactions, and calculated H_2'. We ran 500 bootstrap iterations to calculate standard errors around the estimates of H_2'. The bootstrap procedure was implemented in R 2.9.1 (R Development Core Team 2009), using the bipartite package to obtain estimates of H_2' and its standard errors (Dormann et al. 2009); the bipartite package is *available online*.[8]

To determine the relative importance of a frugivore for a particular plant species, we used the interaction strength metric (Bascompte et al. 2006, Vázquez et al. 2007), a measure of the ecological impact of a species, which differs conceptually from the niche properties H_2' and d_i' because it directly reflects the absolute interaction counts (Blüthgen 2010). Interaction strength is the relative frequency of a frugivore species on a particular plant species (number of interactions between frugivore species j and plant species i divided by number of visits of all frugivore species to plant species i).

To measure the overall impact of a frugivore, we calculated species strength that is the sum of interaction strength values of a frugivore on all plants (Bascompte et al. 2006). Furthermore, to analyze whether interaction strength of frugivores changed with plant height and habitat, we calculated the relative frequency of frugivores of different diet specialization (obligate, partial, opportunistic) and forest dependence (forest species, visitors) on each plant species. First, we tested whether the frequency of visits of obligate, partial, and opportunistic frugivores changed with plant height or forest habitat. Second, we tested whether the frequency of forest species and visitors differed between plant species in different strata or habitats.

Statistical analysis

To investigate plant specialization on frugivores (d_i'), we analyzed the effects of the number of frugivore visitors, plant height, and forest type on d_i'. From the frugivores' perspective, we examined the effects of number of frugivore visits, diet specialization on fruits, and forest dependence on d_j'. We refined comparisons of diet specialization using an orthogonal contrast to compare between fruit specialists (obligate and partial) and opportunists (opportunistic).

To identify properties of frugivores that influence species strength, we analyzed the effects of frugivore frequency in the network, frugivore specialization on specific plants (d_j'), diet specialization on fruits, and forest dependence on species strength. Additive effects of the explanatory variables were tested in multiple regression analyses incorporating diet specialization on fruits and forest dependence as dummy variables. We calculated linear models for all possible combinations of predictor variables ($n = 15$) plus a model including only the intercept. Models were compared according to Akaike's information criterion (AIC). In addition, we calculated the Akaike weight for each model using the R package bbmle 0.9.3 (Bolker 2009), and summed the Akaike weights for each predictor over all models including the respective variable (Burnham and Anderson 2002). The Akaike weight gives the likelihood that a model is the best available model, and thus the summed Akaike weight measures the relative importance of each variable in contributing to the best model; higher weights indicate more important variables (Burnham and Anderson 2002).

Furthermore, we partitioned interaction strength into the contributions of frugivores of different diet specialization and forest dependence, respectively, and tested the effects of plant height and forest type on the respective contributions. To quantify the differences in the frugivore community among forest strata and habitats, we calculated pairwise Bray-Curtis distances, then partitioned differences in community composition among strata and habitats using a MANOVA approach, and tested significance by permuting the raw data (1000 permutations, function adonis in R package Vegan 1.13-1; Oksanen et al. 2008).

All statistical analyses were carried out with R 2.9.1 (R Development Core Team 2009). Prior to analysis, specialization indices (d_i', d_j') were angular-transformed, and plant height, crop mass, the number of frugivore visits to a plant species, the number of visits of a frugivore species to all plant species, and species strength of frugivores were log-transformed. Angular transformation of specialization indices (d_i', d_j') did not completely remove the skewed distribution of model residuals. To account for this uncertainty, we additionally performed all analyses including plant or frugivore

[8] ⟨http://cran.r-project.org/web/packages/bipartite/index.html⟩

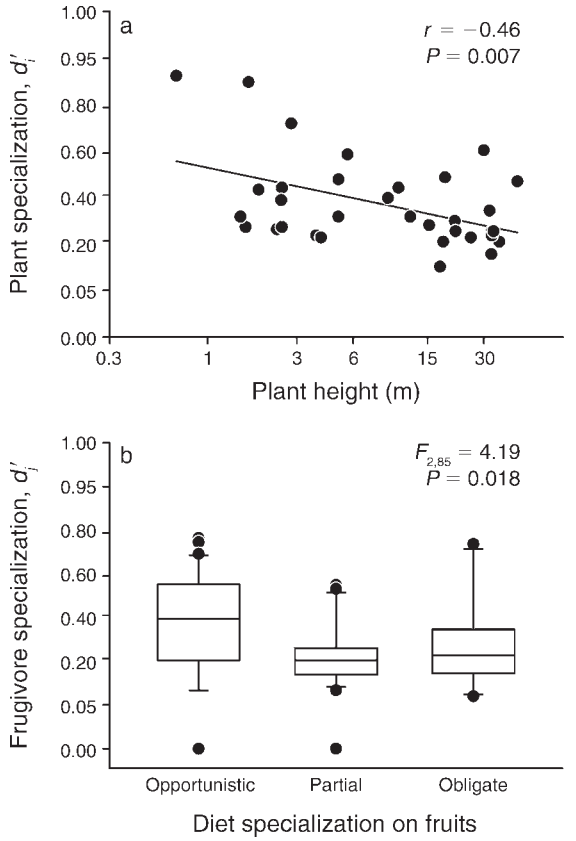


FIG. 1. Specialization of plants on frugivores (d_i') and of frugivores on plants (d_j') in a Kenyan rain forest. Specialization indices (d_i' and d_j') measure the deviation from an expected interaction distribution as standardized Kullback-Leibler distance ranging from 0.0 (extremely generalized) to 1.0 (extremely specialized; Blüthgen et al. 2006). (a) Relationship between d_i' and plant height, and (b) effect of fruit diet specialization (i.e., opportunistic, partial, and obligate frugivores) on d_j'. Note the angular scale for specialization indices and the log scale for plant height. Lines across boxes are medians, boxes indicate 25th and 75th percentiles, whiskers indicate 10th and 90th percentiles, and solid circles are outliers.

specialization (d_i', d_j') with rank-transformed indices. In all analyses, results were the same for angular- and rank-transformed specialization indices.

In all the models described, we tested for phylogenetic effects of plants or frugivores in the residuals of the models (Gittleman and Kot 1990). We calculated Moran's I values of the residuals for each phylogenetic distance class (species pairs differing at the class, order, family, or genus level) for plants and frugivores, respectively (Paradis et al. 2004; the R package ape 2.2-4 is *available online*).[9] Phylogenetic information for plants was obtained from Stevens (2008), for birds from Clements (2008), and for mammals from Kingdon (1997). In all statistical models, Moran's I values did not differ significantly from zero in all phylogenetic distance classes of plants and animals, respectively, indicating that phylogenetic autocorrelation did not distort the results obtained in the models.

[9] ⟨http://cran.r-project.org/web/packages/ape/index.html⟩

RESULTS

Network specialization

Overall, the index of complementary specialization H_2' of the plant–frugivore network was 0.298. Consistent with the stratification hypothesis, specialization was low in the forest canopy ($H_2' = 0.282 \pm 0.007$) [mean ± SE, throughout], whereas specialization was higher in the midstory (0.507 ± 0.029) and understory strata (0.383 ± 0.031). Accordingly, crop mass and frugivore visitation increased with plant height (linear regressions, $n = 33$: $r = 0.84$, $P < 0.001$, and $r = 0.74$, $P < 0.001$, respectively). In contrast to the habitat hypothesis, specialization was similar in primary (0.321 ± 0.008) and secondary forest (0.288 ± 0.021), although crop mass (1292.9 g vs. 151.3 g) and frugivore visitation (68.8 visits vs. 20.6 visits) were higher in primary than in secondary forest (ANOVAs: $F_{1,31} = 6.93$, $P = 0.013$, and $F_{1,31} = 6.38$, $P = 0.017$, respectively).

Specialization of plants and frugivores

Specialization of plants on frugivores (d_i') ranged from 0.112 to 0.907. Specialization decreased with the number of frugivore visitors on a plant (linear regression: $n = 33$, $r = -0.57$, $P < 0.001$), and in line with the stratification hypothesis decreased with plant height (linear regression: $r = -0.46$, $P = 0.007$; Fig. 1a). However, specialization did not differ between plants growing in primary (0.317) and secondary forests (0.419, ANOVA: $F_{1,31} = 2.01$, $P = 0.166$). Specialization of frugivores on plants (d_j') ranged from 0 to 0.781. Specialization was not correlated with the number of visits of a frugivore on fruiting plants (linear regression: $n = 88$, $r = -0.14$, $P = 0.197$). Consistent with the frugivory hypothesis, obligate and partial frugivores were less specialized on specific plant species than opportunistic frugivores (orthogonal contrast in ANOVA: $F_{1,86} = 7.45$, $P = 0.008$; Fig. 1b), but specialization did not differ between forest species (0.259) and visitors (0.329; ANOVA: $F_{1,86} = 2.16$, $P = 0.145$).

Impact of frugivores on plants

Species strength of frugivores on plants ranged from 0.002 to 4.161. The frequency distribution followed a log-normal distribution (KS [Kolmogorov-Smirnov] test: $Z = 0.536$, $P = 0.936$). Species with the strongest impact were Blackcap (*S. atricapilla*, 4.16), Common Bulbul (*P. barbatus*, 3.60), and red-tailed monkey (*Cercopithecus ascanius*, 2.41). Consistent with the abundance and specialization hypothesis, species strength increased with the number of visits of a frugivore (linear regression: $n = 88$, $r = 0.78$, $P < 0.001$; Fig. 2a) and with the degree of specialization on specific plants (linear regression: $n = 88$, $r = 0.39$, $P <$

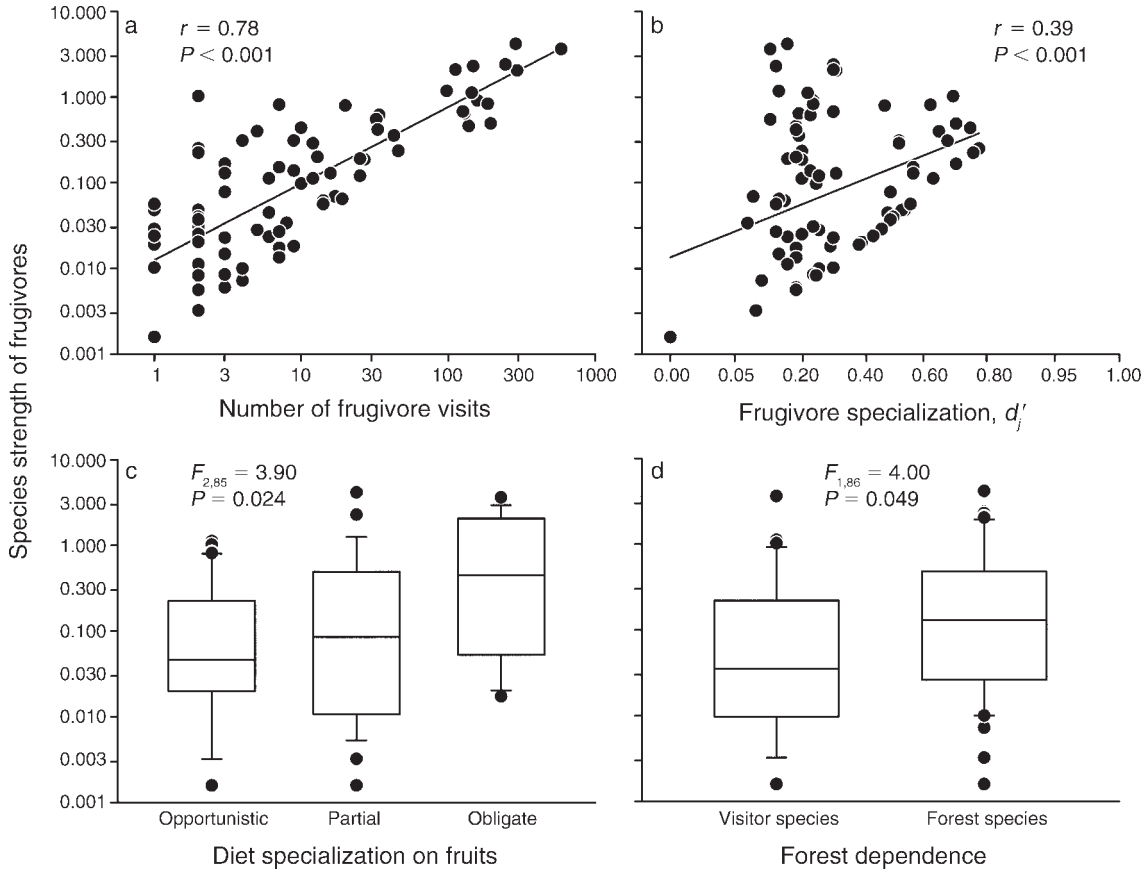


FIG. 2. Species strength of frugivores on plants in relation to (a) number of visits of a frugivore, (b) frugivore specialization on plants (d_j'), (c) degree of diet specialization on fruits, and (d) forest dependence of frugivores, where forest species reproduce almost exclusively in forest, and visitor species usually reproduce in non-forest habitats (for classification details see Bennun et al. 1996). Species strength of frugivores on plants is a relative measure of the overall impact of a frugivore within the network and is calculated by summing the interaction strengths (relative frequencies) of a frugivore species over all plant species (Bascompte et al. 2006). Note the log scale for species strength and number of visits, and the angular scale for specialization index.

0.001; Fig. 2b). Obligate frugivores were more important in terms of species strength than partial and opportunistic frugivores (ANOVA, $F_{2,85} = 3.90$, $P = 0.024$; Fig. 2c), and forest species were more important than visitors (ANOVA, $F_{1,86} = 4.00$, $P = 0.049$; Fig. 2d). This is consistent with the abundance hypothesis because obligate frugivores (mean frequency total per frugivore species = 39.5 individuals) were more common than partial frugivores (11.6 individuals), and partial frugivores were more common than opportunistic frugivores (3.7 individuals; ANOVA, $F_{2,85} = 14.7$, $P < 0.001$), whereas forest species (11.6 individuals) were more numerous than visitors (4.6 individuals; ANOVA, $F_{1,86} = 6.70$, $P = 0.011$).

Comparisons of all multiple models based upon AIC revealed that the best model of species strength included all four predictor variables ($r^2 = 0.87$, $F_{5,82} = 113.1$, $P < 0.001$). All four models that included both the number of visits of a frugivore and the specialization of a frugivore on specific plants were appropriate (model differences in $\Delta AIC < 2$). Akaike weights of all other models were less than 0.001. Thus, species strength of frugivores on plants strongly increased with the number of visits and specialization on specific plants, whereas diet specialization on fruits and forest dependence were much less important (Table 1).

Interaction strength in relation to plant height and forest type

We partitioned the interaction strength of frugivores into the contributions of obligate, partial, and opportunistic frugivores. Plant height had a linear (MANOVA, Pillai trace = 0.47, $F_{2,29} = 13.02$, $P < 0.001$) and a quadratic effect (MANOVA, Pillai trace = 0.25, $F_{2,29} = 4.92$, $P = 0.015$) on the interaction strength of frugivores of different diet specialization, whereas forest type had no effect (MANOVA, Pillai trace = 0.05, $F_{2,30} = 0.79$, $P = 0.463$). The relationship between interaction strength of frugivores and plant height was best described with linear regressions for obligate and opportunistic frugivores, and with a quadratic regression for partial frugivores (Fig. 3). Importantly, forest strata determined

TABLE 1. Effects of the number of visits of a frugivore, specialization on specific plants, forest dependence, and fruit diet specialization on species strength of 88 frugivorous animals in a plant–frugivore network in a Kenyan rain forest.

Source of variation	β	SE	t	P	Akaike weight
Number of visits	0.878	0.0462	19.02	<0.001	1.000
Specialization on plants	0.501	0.0413	12.12	<0.001	1.000
Forest species	0.080	0.0422	1.90	0.060	0.640
Partial frugivores	−0.062	0.0464	−1.34	0.185	0.469
Obligate frugivores	−0.091	0.0477	−1.91	0.060	0.469

Notes: Factorial predictors are represented by dummy variables (i.e., forest species, and partial and obligate frugivores, respectively). According to AIC, the full model was the best model. Given are standardized regression coefficients (β) and their standard errors, and the summed Akaike weights that measure the relative importance of each variable over all multiple models. Overall model: $r^2 = 0.87$, $F_{5,82} = 113.1$, $P < 0.001$.

whether plants interacted with obligate or opportunistic frugivores. As we had expected, canopy plants depended mostly on obligate frugivores, midstory plants on partial frugivores, and understory plants on opportunistic frugivores. Furthermore, we partitioned the interaction strength of frugivores into the contributions of true forest species and visitors. Their contributions did not change with plant height (ANOVA, linear term $F_{1,30} = 0.32$, $P = 0.573$; quadratic term $F_{1,31} = 0.15$, $P = 0.704$) and, in contrast to our expectation, did not differ between plants in primary and secondary forest (ANOVA, $F_{1,31} = 0.46$, $P = 0.503$).

Consistent with the strong effect of forest stratum on interaction strength, the frugivore community differed more strongly among forest strata (function adonis in R package Vegan 1.13-1: $r^2 = 0.172$, $F_{2,30} = 3.12$, $P < 0.001$) than between primary and secondary forest (adonis: $r^2 = 0.066$, $F_{1,31} = 2.18$, $P = 0.019$). Most of the variation among forest strata was due to the contrast between the canopy and the two lower forest layers (adonis: $r^2 = 0.130$, $F_{1,32} = 4.61$, $P < 0.001$).

DISCUSSION

Stratification of plant–frugivore relationships

Overall, specialization in this tropical plant–frugivore network was as low as in other plant–frugivore networks (Blüthgen et al. 2007). Low specialization in plant–frugivore relationships is traditionally expected in tropical ecosystems because the diversity and abundance of plants with fleshy fruits and the year-round dependence of frugivores on fruits are high in tropical forests (Jordano 2000, Kissling et al. 2009). However, our study illustrates that it is important to differentiate among forest strata and between feeding guilds. A very interesting finding of our study is that the degree of generalization in the plant–frugivore network was determined by the vertical stratification of the forest, resulting in generalized plant–frugivore relationships in the canopy and more specialized associations in the mid- and understory. Canopy plants depended mostly on obligate frugivores, midstory plants on partial frugivores, and understory plants on opportunistic frugivores. This might be due to resource-oriented utilization of fruit resources (Levey 1988, Tellería et al. 2008). Canopy plants contributed most to the overall fruit crop (Schaefer et al. 2002), therefore attracting a wide spectrum of obligate frugivores that foraged on many different canopy species with fleshy fruits. In contrast, opportunistic frugivores foraged mostly in lower forest strata where fruit availability and choice were more limited. These findings are consistent with a study of frugivore assemblages on *Ficus* trees where canopy trees attract more and larger-bodied frugivore species than understory trees (Shanahan and Compton 2001).

Generalized fruit choice of obligate frugivores has been reported in previous studies and has been related to seasonal variation in fruit resource abundances (Loiselle and Blake 1990, Carlo et al. 2003, Wallace 2006). In our study, seasonal variation was not sufficient to explain generalization because observations were only carried out in the dry season when fruit diversity and availability were constantly high (Cords 1986; M. Flörchinger and J. Braun, *personal observations*). We rather assume that fruit choice of obligate frugivores is generalized because frugivorous birds frequently move to exploit new fruit resources (Sun et al. 1997) and forage over large distances (Holbrook and Smith 2000), flexibly choosing fruiting plants according to spatial and temporal availability. Another driver of generalization in fruit–frugivore relationships may be resource complementarity (Whelan et al. 1998). A mixed fruit diet provides a higher diversity of nutritional components and may thereby increase bird fitness (Whelan et al. 1998). Although the importance of a mixed diet for tropical frugivores is not known, frugivores from different taxonomic groups (i.e., birds, bats, and monkeys) have all been reported to forage on many different fruit species in tropical forests (Wheelwright 1985, Cords 1986, Wendeln et al. 2000).

In contrast to the pattern in the forest canopy, we found that understory plants and opportunistic frugivores were more specialized on frugivores and fruiting plants, respectively. This pattern was not caused by understory birds feeding only rarely on fruits because the specialization index that we used measures complementary specialization as the deviation from an expected

interaction distribution (Blüthgen et al. 2006). The greater specialization in the understory than in the canopy instead may be caused by lower fruit abundance and diversity in the understory (Schaefer et al. 2002) and by smaller home ranges of understory frugivores (Westcott and Graham 2000), both limiting frugivores' choice. Moreover, the opportunistic frugivores predominant in the understory may be less dependent on a mixed fruit diet because they complement their invertebrate diet with fruits. Morphological differentiation among understory frugivores may also increase differential fruit choice (see Loiselle and Blake 1990).

In contrast to the strong effects of forest stratum, specialization in plant–frugivore relationships did not differ between primary and secondary forest, although fruit availability was higher in primary than in secondary forest. We found a greater differentiation of the frugivore community among strata than between habitats, suggesting that the frugivore community was vertically structured, but hardly differed between forest types. This is probably due to the behavior of frugivores that can track fruits over large distances (Saracco et al. 2004) and frequently pass habitat boundaries in the study area (Eshiamwata et al. 2006, Berens et al. 2008). In contrast, frugivores have been reported to rarely move between forest strata (Malmborg and Willson 1988, Loiselle and Blake 1990).

Determinants of frugivore impact

Ecological communities are composed of many rare and few common species, particularly in tropical ecosystems (McGill et al. 2007). Mostly due to this inherent property of ecological communities, many rare species interact with a few common species and a few common species with many interaction partners (Blüthgen et al. 2008, Vázquez et al. 2009*a*). Accordingly, in our tropical system, species strength of frugivores on plants followed a log-normal distribution. Thus, only a few common frugivore species consumed fruits of many plant species, whereas the majority of frugivores foraged on fruits of specific plants. The three most important frugivores in the network formed a very heterogeneous group: a migratory bird (Blackcap, *Sylvia atricapilla*), a resident bird (Common Bulbul, *Pycnonotus barbatus*), and a monkey (red-tailed monkey, *Cercopithecus ascanius*). In this tropical system, the migratory Blackcap was the most important frugivore, indicating that migratory birds can occupy important ecological positions on their wintering grounds (see also Tellería et al. 2008).

The overall number of frugivore visits was the most important predictor of the impact of a frugivore species. Visitation frequencies of birds were positively correlated with point-count abundances collected at six sites in the same area of the forest (abundance data taken from Farwig et al. [2009], restricted to data from six sites in primary and secondary forest and months from December to March: Spearman rank $\rho = 0.42$, $P < 0.001$, $n = 83$ bird species). Moreover, we found that the impact of frugivores on fruit plants increased with the degree of specialization on fruits and with frugivores' forest dependence. These effects can be explained by the high visitation frequencies of obligate frugivores and true forest species, respectively. Our findings are consistent with previous studies stressing that species abundance and resource availability are crucial determinants of interaction strength (Vázquez et al. 2009*a*, *b*). Such patterns suggest that common species may be particu-

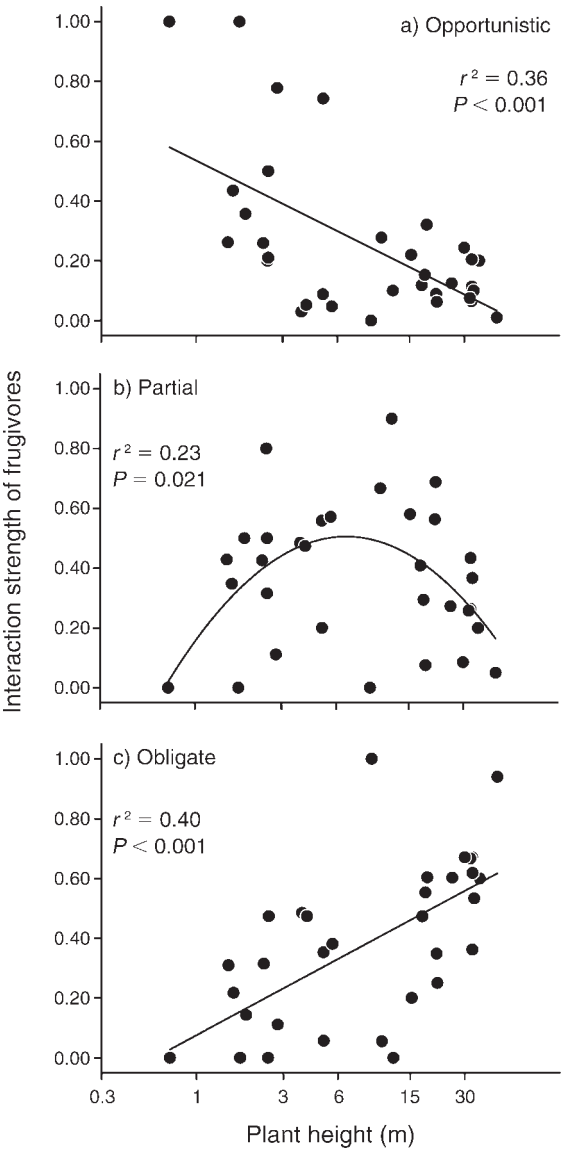


FIG. 3. Relationships between plant height and the interaction strength of frugivores with different degrees of diet specialization: (a) opportunistic, (b) partial, and (c) obligate frugivores. Interaction strength is the relative frequency of a frugivore species on a particular plant species (number of interactions between frugivore species j and plant species i divided by number of visits of all frugivore species to plant species i). Note the log scale for plant height.

larly important for ecosystem processes (Vázquez et al. 2005, Carnicer et al. 2009).

However, in our system, the impact of a frugivore was also affected by specialization on particular plant species. Thus, not only abundant and generalized frugivores matter but also species that are rare, but specialized on specific plants. This suggests that plant–frugivore interactions are not only shaped by neutral abundance effects but also by the foraging niches of frugivores, at least in this tropical system. In contrast, a study from a temperate system did not find strong ecological effects on the interaction strength of frugivores (Carnicer et al. 2009).

Consequences for seed-dispersal services

We recorded fruit removal by vertebrates on plants with fleshy fruits, but lack information on the ultimate seed-dispersal services provided by the frugivores. Nevertheless, our observations suggest that many frugivores dispersed the seeds of their food plants because most fruits were swallowed. In particular, the many medium-to-large frugivores observed in this study, including monkeys, hornbills, and turacos, have gut passage times of several hours and large home ranges (Sun et al. 1997, Yumoto et al. 1999, Holbrook and Smith 2000). Hence, they probably disperse seeds away from their food plants. Moreover, a previous study found that the number of visits and departures of frugivores is more important for the effect of a seed disperser on a plant than the number of fruits dispersed per visit (Vasquez et al. 2005). Thus, our findings present a valuable starting point to assess seed-dispersal services between different groups of frugivores and in different strata of tropical forests.

Our results suggest that obligate frugivores provide reliable seed-dispersal services for canopy trees owing to their high abundance. Generalized fruit choice of obligate frugivores could mitigate spatiotemporal variation in the abundance of consumer species and thereby increase the robustness of seed-dispersal interactions. Similar patterns have been found for pollination networks (Memmot et al. 2004). Generalization might also increase the quality of seed-dispersal services. From seed-dispersal theory, generalization of plant–frugivore relationships is expected to have an advantage for plants when the effects of frugivores on seed dispersal are complementary (Fleming et al. 1993). Complementary effects of frugivores include differences in microhabitat use and in the distances over which seeds are carried away from the food plant (Jordano et al. 2007). Such effects are likely in species-rich frugivore communities, such as tropical forests (Kissling et al. 2009), that span a wide range of body sizes and behavioral traits. Our results suggest that the higher species richness and generalization in the canopy result in more complex dispersal kernels in canopy than in understory plants. Similarly, a higher probability of seed dispersal in the canopy than in the understory has been suggested for a Bornean *Ficus* community (Shanahan and Compton 2001).

In the lower forest strata, plants appear to depend on seed-dispersal services by rare, specialized, and opportunistic frugivores, suggesting lower reliability of seed-dispersal services. Moreover, small understory frugivores have short times of gut passage (10–20 minutes), limiting seed dispersal to rather short distances of ~50 m (Westcott and Graham 2000). On the one hand, seed dispersal may be less important for tropical understory plants because species with fleshy fruits often propagate clonally (Schleuning et al. 2008). On the other hand, understory plants depend on efficient seed dispersal to escape from competition with clonal propagules and to colonize new sites (Schleuning et al. 2009). Thus, the opportunistic frugivores that were crucial frugivores in the understory occupy an important ecological role for plants growing in the lower strata of tropical forests.

At least in this African rain forest, the understory stratum of the plant–frugivore network exhibited low redundancy and was built upon opportunistic frugivores that switch resources according to resource availability (see Carnicer et al. 2009). Sensitivity of seed dispersal to human disturbance has been reported in a study of an East African midstory tree that received low visitation rates from a small disperser community (Cordeiro and Howe 2003). In contrast to canopy trees that responded little and idiosyncratically to local disturbance and habitat fragmentation (Farwig et al. 2006, Kirika et al. 2008*a*, *b*), seed removal in the midstory tree species was drastically reduced in habitat fragments (Cordeiro and Howe 2003). Our results also suggest that seed dispersal in the under- and midstory of tropical forests is less buffered against species loss because it depends on fewer and more specialized species.

Conclusions

In this Kenyan rain forest, we found contrasting patterns in plant–frugivore relationships in different forest strata. In the canopy, obligate frugivores were abundant and foraged on a wide array of plant species. Due to their high interaction frequency and mobility, obligate frugivores are key species for the seed dispersal of canopy trees, at least in our system. In contrast, opportunistic frugivores were predominant in the understory and were rather specific in fruit choice, possibly due to their small home ranges. The strong differentiation between forest strata is at least partly driven by the stratification of fruit resources, which contributes to feeding guild composition. The vertical stratification of the frugivore community into obligate and opportunistic feeding guilds has important consequences for the structure of the plant–frugivore network. In the canopy, where fruits are abundant, the network comprises strong links and generalized associations, whereas in the understory, where fruits are scarce, the network is composed of weak links and more specialized associations. Consequently, our results sug-

gest that seed-dispersal relationships of plants in lower forest strata are more prone to disruption than those of canopy trees.

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APPENDIX

Interaction frequencies in a plant–frugivore network of 88 frugivore and 33 plant species from Kakamega Forest, Kenya (*Ecological Archives* E092-003-A1).